NTNU
Kunnskap for en bedre verden

IDATG2901 - Bachelor task

Main report

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Abstract

Lorem Ipsum is simply dummy text of the printing and typesetting industry. Lorem Ipsum has been the industry's standard dummy text ever since the 1500s, when an unknown printer took a galley of type and scrambled it to make a type specimen book. It has survived not only five centuries, but also the leap into electronic typesetting, remaining essentially unchanged. It was popularised in the 1960s with the release of Letraset sheets containing Lorem Ipsum passages, and more recently with desktop publishing software like Aldus PageMaker including versions of Lorem Ipsum.

Sammendrag

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Forord

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We would like to thank [Supervisor name] for guidance and feedback throughout the project, and [External contact] for providing domain insights and access to relevant resources. We also thank [any other contributors] for their support.

Gjøvik, 13. februar 2026

[Name 1], [Name 2], [Name 3]

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proof of concept backend frontend

1 Innledning

Denne prosjektplanen legger grunnlaget for hvordan gruppens bacheloroppgave skal gjennomføres. Den beskriver prosjektets mål og hvordan gruppen skal samarbeide for å nå disse. Planen redegjør for målsettinger, omfang, organisering, metode og tilnærming til kvalitetssikring, og fungerer som et styrende dokument for planlegging, gjennomføring og evaluering gjennom hele prosjektperioden. Planen inkluderer også en gruppekontrakt som definerer regler, ansvar og forventninger knyttet til samarbeidet i gruppen.

1.1 Bakgrunn

Helsedirektoratets nettsider inneholder store mengder informasjon rettet mot et bredt spekter av målgrupper, inkludert helsepersonell, kommuner, forskere og allmennheten. Til tross for at denne informasjonen er tilgjengelig, opplever brukere ofte utfordringer med å finne relevant innhold, selv når det faktisk finnes. Manglende effektiv gjenfinnbarhet skaper barrierer for god informasjonsinnhenting og kan bidra til redusert tillit til offentlige digitale tjenester.

1.2 Prosjektbeskrivelse fra oppgavebeskrivelsen

Denne bacheloroppgaven tar utgangspunkt i en oppgavebeskrivelse gitt av Helsedirektoratet, med mål om å undersøke hvordan søk og navigasjon på helsedirektoratet.no kan forbedres ved hjelp av kunstig intelligens.

Oppgaven innebærer å kartlegge eksisterende søkefunksjonalitet og navigasjonsstruktur, identifisere svakheter og vurdere hvordan ny teknologi, særlig kunstig intelligens (KI), kan bidra til en forbedret brukeropplevelse. Dette inkluderer blant annet vurdering av KI-drevet søk, automatisk kategorisering og tagging av innhold, personalisering og analyse av brukerdata.

Målet med prosjektet er å utvikle og evaluere en prototype eller et konsept som demonstrerer hvordan KI kan bidra til mer effektiv og relevant informasjonsinnhenting og navigasjon på helsedirektoratet.no.

1.3 Mål og leveranser

Det overordnede målet med prosjektet er å undersøke hvordan kunstig intelligens kan bidra til forbedret søk og navigasjon på helsedirektoratet.no, og å demonstrere dette gjennom utvikling og evaluering av et konkret konsept eller en prototype.

For å oppnå dette er følgende delmål definert:

- Kartlegge og analysere eksisterende søkefunksjonalitet og navigasjonsstruktur på helsedirektoratet.no.
- Identifisere sentrale utfordringer knyttet til informasjonsgjenfinning for ulike brukergrupper gjennom brukerundersøkelser.
- Vurdere relevante KI-baserte teknologier, herunder semantisk søk og automatisk kategorisering og tagging av innhold.
- Utforme en prototype eller et konsept som adresserer identifiserte utfordringer innen søk og navigasjon.
- Evaluere løsningen gjennom brukerundersøkelser med en tilsvarende brukergruppe for å vurdere om løsningen representerer en forbedring sammenlignet med eksisterende løsning.

I prosjektet skilles det mellom søk og navigasjon, der søk i hovedsak forbedres gjennom KI-baserte og backend-drevne løsninger, mens navigasjon primært optimaliseres gjennom brukerinnsikt og manuelle justeringer av informasjonsstruktur og grensesnitt. Samtidig anerkjennes det at forbedringer i søk også kan bidra til en mer helhetlig og effektiv navigasjonsopplevelse.

Prosjektets leveranser består av en skriftlig bachelorrapport som dokumenterer arbeidet, samt en prototype eller konseptuell løsning med tilhørende dokumentasjon og evaluering.

1.3.1 Resultatmål

Resultatmålene beskriver de konkrete og etterprøvbare resultatene prosjektet skal levere. Ved fullført prosjekt forventes følgende resultater:

- En analyse av eksisterende søkefunksjonalitet og navigasjonsstruktur på helsedirektoratet.no, inkludert identifiserte styrker og svakheter.
- Et konseptuelt design eller en prototype som demonstrerer hvordan kunstig intelligens kan anvendes for å forbedre søk og navigasjon.
- En evaluering av den foreslåtte løsningen basert på relevans, brukervennlighet og tilpasningsevne, støttet av brukerundersøkelser eller simulerte bruksscenarier.
- En skriftlig bacheloroppgave som dokumenterer metodikk, designvalg, evalueringsresultater og etiske vurderinger.

1.3.2 Effektmål

Effektmålene beskriver hvilken verdi prosjektets resultater kan bidra med for brukere og interessenter. Prosjektet har som mål å bidra til følgende effekter:

- Forbedret relevans i søkeresultater, vurdert gjennom brukertesting og relevansvurderinger.
- Redusert gjennomsnittlig tid brukt på å finne spesifikk informasjon i brukertester.
- Økt opplevd brukervennlighet og brukertilfredshet, målt gjennom spørreskjema etter testing.
- Økt transparens og konsistens i organiseringen av innhold gjennom bruk av automatisert klassifisering og tagging.

1.3.3 Rammer

Prosjektet gjennomføres som en bacheloroppgave innenfor rammene av et studieprogram på 22,5 studiepoeng. Den tilgjengelige arbeidsmengden tilsvarer om lag 30 arbeidstimer per student per uke over en periode på cirka 18 uker, noe som gir en samlet arbeidsinnsats på omtrent 540 timer per student. For prosjektgruppen på tre studenter utgjør dette totalt rundt 1 620 arbeidstimer.

Innenfor disse rammene legges det vekt på konseptuell utforming, prototyping og evaluering, fremfor fullskala utvikling eller produksjonssetting. Den foreslåtte løsningen skal derfor betraktes som et proof of concept, med hensikt å demonstrere gjennomførbarhet og potensiell nytteverdi av KI-baserte tilnærminger.

Prosjektet er strukturert rundt utviklingen av en prototype for en webbasert løsning som etterligner utvalgte funksjoner fra *helsedirektoratet.no*. Prototypen benytter tilgjengelige offentlige Application interface-er (API-er) for å hente domeneinformasjon, og utvides med en eksperimentell, KI-støttet søk- og navigasjonsmekanisme. Målet er ikke å erstatte eksisterende løsning, men å utforske alternative samhandlings- og informasjonsgjenfinningsstrategier i et kontrollert testmiljø.

1.3.4 Avgrensninger

Prosjektet er avgrenset av både omfang, tidsrammer og tilgang på ressurser. Gruppen har ikke hatt tilgang til intern infrastruktur, kildekode eller produksjonsmiljøer fra helsedirektoratet.no. Løsningen som utvikles baserer seg derfor på offentlig tilgjengelig informasjon og åpne API-er, og implementeres som en selvstendig prototype utviklet fra bunnen av.

For å sikre gjennomførbarhet innenfor prosjektets rammer er arbeidet avgrenset til funksjonalitet som direkte påvirker søk og navigasjon for brukere. Innhold og seksjoner som ikke har vesentlig relevans for informasjonsgjenfinning, slik som nyhetsseksjoner, arrangementer, konferanser og øvrig redaksjonelt innhold (for eksempel *Aktuelt* og *Arrangementer*), inngår derfor ikke i prototypen.

Videre er prosjektet avgrenset til et utvalg tematiske sider (temasider) som benyttes som representative eksempler for å utforske og demonstrere forbedret søk og navigasjon. Disse temasidestrukturene er utformet av prosjektgruppen og reflekterer ikke nødvendigvis full bredde eller detaljnivå i eksisterende løsninger hos Helsedirektoratet.

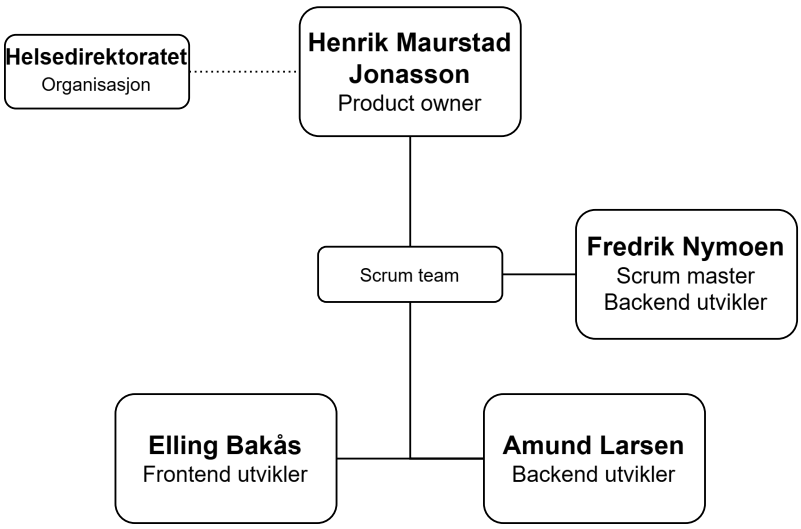
Ettersom prosjektet ikke har tilgang til en etablert brukerbase, er innsamling av reelle brukerdata begrenset. Brukerundersøkelser og evaluering gjennomføres derfor på et begrenset antall testbrukere, og resultatene må tolkes innenfor denne konteksten. Prosjektet har heller ikke som mål å utvikle en fullstendig produksjonsklar frontend- og backend-løsning, men fokuserer på konseptuell utforming, prototyping og evaluering av forbedrede søke- og navigasjonsmekanismer.

1.4 Gruppens bakgrunn

Prosjektet er organisert med tydelige roller, ansvarsområder og samarbeidsrutiner for å sikre strukturert og effektiv gjennomføring av arbeidet. Gruppen består av tre studenter med komplementær faglig kompetanse innen programvareutvikling, og arbeidet er organisert etter prinsipper fra smidig utviklingsmetodikk.

1.4.1 Organisering

En oversikt over prosjektgruppens organisering og rollefordeling er illustrert i Figur 1.



Figur 1: Oversikt over prosjektgruppens struktur og rollefordeling.

1.4.2 Prosjektroller

Prosjektgruppen er organisert etter etablerte roller inspirert av Scrum-rammeverket, med en tydelig fordeling av ansvar:

- **Scrum master / prosjektleder: Fredrik Nymoen**
  - Overordnet ansvar for fremdrift og koordinering av prosjektet.
  - Kontaktpunkt mot produktansvarlig (product owner).
  - Planlegging og gjennomføring av møter.
  - Sikre rettferdig fordeling av arbeidsoppgaver i gruppen.
  - Tilrettelegging for smidige arbeidsprosesser, inkludert sprintplanlegging og oppfølging.
- **Utvikler: Amund Larsen**
  - Ansvar for backend-utvikling.
  - Ansvar for struktur, vedlikehold og kvalitetssikring av hovedrapporten i Overleaf, inkludert språklig og faglig konsistens.
  - Bidrar til å sikre at prosjektets leveranser følger avtalte standarder og er i tråd med gruppekontrakten.
- **Utvikler: Elling Bakås**
  - Ansvar for frontend-utvikling.
  - Ansvar for referatføring fra formelle møter, inkludert møter med produktansvarlig, veileder, sprintplanlegging og sprintgjennomganger.
- **Product owner: Henrik Maurstad Jonasson**
  - Representerer Helsedirektoratet som oppdragsgiver.
  - Formidler oppdragsgivers behov, forventninger og visjon for prosjektet.
  - Bidrar med innspill til prioritering av funksjonalitet og innhold i prosjektets produktbacklog.

1.4.3 Faglige ansvarsområder

De faglige ansvarsområdene i prosjektet er fordelt i tråd med gruppemedlemmenes kompetanse og erfaring. Dette legger til rette for effektiv arbeidsdeling, samtidig som gruppen samarbeider tett på tvers av roller for å sikre helhetlige og konsistente løsninger.

1.5 Rapportens oppbygning

Kapittel X beskriver prosjektets organisering og roller, mens kapittel Y redegjør for metodisk tilnærming.

Det gir ekstra strukturpoeng.

2 Kravspesifikasjon

2.1 Funksjonelle krav

2.1.1 Brukermønstre

2.1.2 Domenemodell

2.2 Ikke-funksjonelle krav

2.3 Driftskrav

2.3.1 Åpen kildekode

3 Utviklingsprosess

Prosjektplanen, presentert i vedlegg Kapittel B, beskriver hvordan gjennomføringen av prosjektet opprinnelig var planlagt. Dette kapittelet bygger videre på planen og gir en redegjørelse for hvordan prosjektet faktisk ble gjennomført i praksis. Selv om den overordnede metodikken ble fulgt, ble enkelte elementer justert underveis for å tilpasse arbeidsform, fremdrift og samarbeid med eksterne aktører.

3.1 Møtestruktur og koordinering

Daglige statusmøter (Daily Scrum)

Daglige statusmøter ble gjennomført kl. 09:15 fra mandag til fredag, primært fysisk når det var mulig. Møtene hadde kort varighet og fulgte en fast struktur hvor hvert medlem redegjorde for:

1. Hva som var gjennomført siden forrige møte
2. Hva som skulle arbeides med videre
3. Eventuelle hindringer eller utfordringer

Denne strukturen bidro til kontinuerlig koordinering og transparens i arbeidsprosessen. Eventuelle tekniske eller organisatoriske utfordringer ble identifisert tidlig, noe som reduserte risiko for forsinkelser og bidro til stabil fremdrift gjennom sprintene.

Veiledermøter

Det ble gjennomført ukentlige veiledermøter hver onsdag fra kl. 12:00 til 13:00. Disse møtene hadde en mer strukturert og refleksiv karakter enn de daglige statusmøtene. Fokus lå på fremdrift i prosjektet, kvaliteten på metodiske valg, samt diskusjon av forskningsmessige utfordringer.

Veiledermøtene ble også brukt aktivt til å få tilbakemeldinger på rapportstruktur, argumentasjon og akademisk presisjon. Dette bidro til å sikre metodologisk konsistens og faglig forankring gjennom hele prosjektperioden, samtidig som det ga rom for kritisk refleksjon rundt både løsning og forskningsdesign.

Sprintgjennomgang (Sprint Review)

Ved slutten av hver sprint ble det gjennomført sprintgjennomgang annenhver mandag kl. 10:30 sammen med Product Owner. I disse møtene ble ferdigstilt funksjonalitet demonstrert og vurdert opp mot definerte sprintmål.

I tillegg til selve demonstrasjonen ble det diskutert teknologiske løsninger, metodiske valg og arkitektoniske vurderinger. Det ble gjort sammenligninger mellom prosjektets tilnærming og Helsedirektoratets egne systemer og pågående initiativer. Dette bidro til å holde prosjektgruppen oppdatert på relevante interne føringer og teknologiske rammebetingelser hos oppdragsgiver.

Sprintgjennomgangene fungerte dermed ikke kun som en kontrollmekanisme for fremdrift, men også som en faglig dialogarena. Gjennom disse møtene fikk gruppen verdifulle referansepunkter for videre utvikling og spesialisering av løsningen, samt grunnlag for justering av prioriteringer i påfølgende sprint.

Sprintretrospektiv

Etter hver sprintgjennomgang ble det gjennomført en intern sprintretrospektiv i Scrum-teamet. Formålet med retrospektivet var å evaluere samarbeidsprosesser, arbeidsflyt og effektivitet i gjennomføringen av sprinten.

Retrospektivene resulterte i konkrete forbedringstiltak, blant annet tydeligere avgrensning av sprintmål, mer presis oppgavefordeling og forbedret dokumentasjonspraksis. Denne iterative forbedringsprosessen bidro til økt modenhet i teamets arbeidsform over tid.

Utvidede demonstrasjonsmøter med Helsedirektoratet

I tillegg til ordinære sprintgjennomganger ble det annenhver sprint arrangert en utvidet demonstrasjon med Helsedirektoratet, hvor flere interne representanter deltok, ikke kun Product Owner. Disse møtene hadde en bredere organisatorisk forankring og ga innsikt i hvordan løsningen ble oppfattet av ulike fagmiljøer.

De utvidede demonstrasjonene bidro til tverrfaglige tilbakemeldinger og styrket interessentforankringen i prosjektet. Ved å eksponere løsningen for flere perspektiver ble både tekniske og faglige implikasjoner tydeligere, noe som igjen påvirket videre prioriteringer og utviklingsretning. Dette styrket prosjektets relevans og tilpasning til faktiske behov hos oppdragsgiver.

3.2 Utviklingsmodell? / Scrumming

3.2.1 Gjennomføring av sprint

3.2.2 Sammendrag av iterasjoner

05.01

3.3 Erfaringer og refleksjon

4 Metode

Dette kapittelet beskriver hvordan datagrunnlaget for oppgaven er samlet inn, hvilke metoder som er benyttet, samt hvordan utvalget er rekruttert og distribuert geografisk. Formålet med datainnsamlingen har vært å få innsikt i brukeres erfaringer med søk og navigasjon på helsedirektoratet.no, samt identifisere forbedringsområder.

4.1 Datainnsamling

Datainnsamlingen er gjennomført ved hjelp av en kvantitativ spørreundersøkelse utviklet i Microsoft Forms[1]. Undersøkelsen er vedlagt i Kapittel A og inneholder både lukkede og åpne spørsmål.

Spørreskjemaet omfatter blant annet:

- Brukerens rolle/kategori (f.eks. helsepersonell, offentlig forvaltning, privatperson).
- Bruksfrekvens av helsedirektoratet.no.
- Formål med besøket.
- Vurdering av søkefunksjon og navigasjon (skala 1–10).
- Opplevde utfordringer ved søk.
- Tiltak brukeren gjør dersom informasjon ikke finnes.
- Forslag til forbedringsområder.

Spørsmålene er utformet for å gi både målbare vurderinger (kvantitative data) og kvalitative beskrivelser av brukeropplevelse. Selve spørreskjemaet er dokumentert i vedlegg Kapittel A Undersøkelsen er anonym, og det er ikke samlet inn personidentifiserende informasjon.

4.2 Rekruttering av respondenter

Opprinnelig var planen å publisere undersøkelsen på Helsedirektoratets egne nettsider for å nå et bredt spekter av brukere. Dette lot seg imidlertid ikke gjennomføre innenfor prosjektperioden.

Som alternativ distribusjonsstrategi ble undersøkelsen:

- Delt internt via Helsedirektoratets kommunikasjonskanaler.
- Sendt direkte til 11 sykehus i Norge via e-post.
- Distribuert gjennom relevante faglige og personlige nettverk til brukere av helsedirektoratet.no.

Denne tilnærmingen har gitt tilgang til både interne og eksterne brukere av nettsiden, inkludert helsepersonell, ansatte i offentlig sektor og privatpersoner.

4.3 Lokal datainnsamling

For å styrke datagrunnlaget ytterligere ble det også tatt direkte kontakt med lokalt helsepersonell. Det ble blant annet etablert kontakt med representanter ved Sykehuset Innlandet, Gjøvik, som videreformidlet undersøkelsen internt.

Det ble også forsøkt å etablere kontakt med Institutt for helsevitenskap i Gjøvik for å nå relevante fagmiljøer innen utdanning og forskning.

4.4 Geografisk spredning

Selv om undersøkelsen er anonym, er det kjent at respondenter har deltatt fra følgende fylker:

[1] https://forms.office.com

- Nordland
- Oslo
- Trøndelag
- Innlandet
- Vestland

Dette gir en viss geografisk spredning og bidrar til et mer representativt datagrunnlag på tvers av ulike regioner og brukergrupper i Norge.

4.5 Sekundærdata

I tillegg til egen datainnsamling har vi fått tilgang til tidligere brukerundersøkelser gjennomført av Helsedirektoratet. Disse fungerer som sekundærdata og gir et sammenligningsgrunnlag for å identifisere mønstre og vedvarende utfordringer knyttet til søk og navigasjon.

Kombinasjonen av primærdata (egen undersøkelse) og sekundærdata styrker analysens validitet ved at funn kan sees i sammenheng med tidligere målinger.

4.6 Git workflow (praksis)

4.7 QA -Kvalitetssikring

4.7.1 Kode praksiser

5 Design og implementasjoner

struktur og oversikt over vår løsning.

Kode 1: Example script

```
print("Hello, world!")
```

6 Teori og teknologier

7 Dagens løsning

Hvordan helsedirektoratet.no fungerer i dag

7.1 Brukerundersøkelser

7.1.1 Oppsummering av data (pp/excel) /brukerundersøkelser

tilbakemeldinger og forbedring

8 Søk

Søk er backend løsning og hvordan vi går fram der med vår løsning

8.1 Database

9 Navigasjon

Navigasjon er mer en frontendløsning, vår løsing of framgang.

10 Trening av ki modeller

10.1 Semantic search

10.2 Embedding

10.3 Elastic search

10.4 Key-word search

10.5 Sammensetning av søk optimalisering

11 Vurdering av løsning

12 Diskusjon

13 Konklusjon

Vedlegg A

Brukerundersøkelser

Søk og navigasjon på helsedirektoratet.no

Vi gjennomfører en kort undersøkelse om søk og navigasjon på helsedirektoratet.no som del av en bacheloroppgave i samarbeid med Helsedirektoratet.

Undersøkelsen tar under 5 minutter, og svarene behandles anonymt.

* Obligatorisk

1. Hvilken kategori jobber du innenfor / hører du til? *

- ◯ Helsepersonell
- ◯ Ansatt i Helsedirektoratet
- ◯ Privatperson
- ◯ Utdanning / forskning
- ◯ Kommune / Offentlig forvaltning
- ◯ Privat aktør
- ◯ Annet

2. Hvor ofte bruker du helsedirektoratet.no? *

- ◯ Dagentlig
- ◯ Ukentlig
- ◯ Månedlig
- ◯ Sjeldnere enn månedlig
- ◯ Første gang
- ◯ Husker ikke / Vet ikke

3. Hvorfor besøker du helsedirektoratet.no? (skriv 1-2 setninger)

4. Hvor fornøyd er du totalt sett med søk og navigasjon på helsedirektoratet.no? (1 stjerne = veldig misfornøyd, 10 stjerner = veldig fornøyd)

☆ ☆ ☆ ☆ ☆ ☆ ☆ ☆ ☆ ☆

5. Hvor lett er det å finne informasjonen du leter etter? (1 stjerne = veldig vanskelig, 10 stjerner = veldig lett)

☆ ☆ ☆ ☆ ☆ ☆ ☆ ☆ ☆ ☆

6. Hvordan vil du kort oppsummere din opplevelse av søk og navigasjon på helsedirektoratet? (skriv 1-2 setninger)

7. Hva opplever du som største utfordring med søk? (kan velge flere) *

- [ ] Treffer ikke det jeg leter etter
- [ ] For mange / irrelevante treff
- [ ] Vanskelig å forstå hva jeg skal søke på
- [ ] Søkeresultatene er uoversiktlige
- [ ] Har ingen spesielle utfordringer

8. Hvordan opplever du navigasjonen (menyer, kategorier, struktur)? (1 stjerne = veldig misfornøyd, 10 stjerner = veldig fornøyd)

☆ ☆ ☆ ☆ ☆ ☆ ☆ ☆ ☆ ☆

9. Når du ikke finner det du leter etter, hva gjør du oftest?

- ◯ Har ikke hatt dette problemet, finner alltid frem
- ◯ Fortsetter å søke til jeg finner det jeg leter etter på helsedirektoratet.no
- ◯ Bruker andre kilder enn helsedirektoratet.no (F.eks ChatGPT, Google osv)
- ◯ Avslutter søket / prøver senere
- ◯ Annet

10. Hva mener du er det/de viktigste forbedringsområdene?

- [ ] Mer treffsikkert søk – Søkeresultatene viser oftere relevant innhold og riktige sider.
- [ ] Mer logisk struktur og navigasjon – Innholdet er mer intuitivt organisert og lettere å finne.
- [ ] Bedre filtrering og sortering – Enklere å avgrense innhold etter tema, målgruppe, dato eller dokumenttype.
- [ ] Veiledning eller digital assistent (f.eks. chatbot) – Hjelp til å finne riktig informasjon ved å stille spørsmål i naturlig språk.
- [ ] Bedre oversikt på landingssider – Viktig og ofte brukt informasjon er lettere tilgjengelig med færre klikk.
- [ ] Personalisering av innhold – Mer relevant innhold basert på rolle, interesser eller tidligere bruk.
- [ ] Vet ikke

11. Hva fungerer godt i dag - og hva kan bli bedre? (skriv 1-3 setninger)

Dette innholdet er verken opprettet eller godkjent av Microsoft. Dataene du sender, sendes til skjemaeieren.

Microsoft Forms

Vedlegg B

Prosjekt plan

NTNU
Kunnskap for en bedre verden

IDATG2901 - Bachelor task

Project plan

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1 Introduction

This project plan lays the foundation for how the group's bachelor thesis work will be carried out. It describes the project's goals and how the group will collaborate to achieve them. The plan outlines the objectives, scope, organization, methodology, and quality assurance approach, and serves as a guiding document for planning, execution, and evaluation throughout the period. It also includes a group contract that defines the rules, responsibilities, and expectations governing the group's collaboration.

2 Objectives and Framework

This section outlines the context, objectives, and methodological framework that define the scope and direction of the bachelor project.

2.1 Background

Helsedirektoratet's websites contain a substantial amount of information intended for a wide range of target groups, including healthcare professionals, municipalities, researchers, and the general public. Despite the availability of this information, users often experience difficulties locating relevant content, even when it exists. This lack of efficient discoverability creates barriers to effective information retrieval and may ultimately reduce user confidence in public digital services.

2.2 Project Goals

The primary goal of this bachelor project is to investigate how artificial intelligence (AI) can be used to improve search functionality and navigation on helsedirektoratet.no. The project aims to analyze the current search and navigational structure, identify key weaknesses, and explore how AI-based techniques can address these challenges.

Furthermore, the project aims to design and evaluate a prototype or conceptual solution that illustrates how AI-driven approaches, including semantic search, personalization, and automated content categorization, may enhance user experience and enable more effective information retrieval.

Impact Goals

The intended effects of the project are formulated to be assessable through qualitative and quantitative evaluation. The project aims to achieve the following effects:

- A measurable improvement in search relevance, evaluated through user testing and relevance assessments.
- A reduction in the average time required for users to locate specific information during usability tests.
- Improved perceived usability and user satisfaction, measured through post-test questionnaires.
- Increased transparency and consistency in content organization through the use of automated classification and tagging methods.

Performance goals

The result goals define the concrete outcomes and deliverables of the project. Upon completion, the project is expected to produce the following results:

- An analysis of the existing search functionality and navigational structure on helsedirektoratet.no, including identified strengths and weaknesses.
- A conceptual design or prototype demonstrating how artificial intelligence can be applied to improve search and navigation.
- An evaluation of the proposed solution based on relevance, usability, and adaptability, supported by user testing or simulated use cases.
- A written bachelor thesis documenting the methodology, design choices, evaluation results, and ethical considerations.

Learning goals

The learning goals describe the competencies and knowledge the students aim to acquire through the project. Upon completion, the students are expected to have:

- Acquired hands-on experience with the SCRUM framework, including sprint planning, iterative development, and continuous improvement.
- Gained practical experience in analyzing and improving complex information systems.
- Developed an understanding of AI-based technologies relevant to search, natural language processing, and personalization.
- Acquired experience in designing, implementing, and evaluating a prototype solution.
- Strengthened skills in user-centered design, usability testing, and data-driven evaluation.
- Improved academic writing and documentation skills through the completion of a structured bachelor thesis.

2.3 Framework

This project is conducted as a bachelor thesis within the scope of a 22.5 ECTS study program. The allocated workload corresponds to approximately 30 hours of work per student per week over an estimated period of 18 weeks, resulting in a total workload of approximately 540 hours per student. For the three-person project group, this amounts to an overall effort of approximately 1 620 person-hours.

Given these constraints, the project emphasizes conceptual design, prototyping, and evaluation rather than full-scale development or deployment in a production environment. The proposed solution is therefore positioned as a proof of concept, intended to demonstrate the feasibility and potential value of AI-based approaches rather than to deliver a fully operational system.

The project is structured around the development of a prototype web solution that emulates selected functionality of *helsedirektoratet.no*. The prototype consumes existing public APIs to retrieve domain-specific information and extends this foundation with an experimental AI-assisted search and navigation mechanism. The objective is not to replicate or replace the existing platform, but to explore alternative interaction patterns and information retrieval strategies in a controlled and test-oriented environment.

3 Project Scope

This section defines the problem domain, scope limitations, and envisioned product, clarifying what the project addresses and what lies outside its boundaries.

3.1 Problem area

Helsedirektoratet is a government agency responsible for developing, implementing, and providing guidance on national health and care policies in Norway. Their website contains large amounts of important information such as clinical guidelines, regulations, recommendations, statistics, and resources for health professionals, municipalities, researchers, and the general public [1].

Because the website serves many different user groups and contains a very large and diverse set of documents, users often experience difficulties finding relevant information, even when it exists. The current search and navigation structures are primarily keyword-based and hierarchical, which makes it challenging to locate content when users do not know the exact terminology, document type, or where the information is placed.

This is problematic because the website is a critical channel for communicating authoritative health information. When users cannot easily find what they are looking for it can lead to misinformation or misinterpretation of rules and recommendations. Improving search and navigation is therefore essential to ensure effective information access, support evidence-based decision-making, and strengthen the usability of the website.

3.2 Problem Limitations

This project develops a prototype that demonstrates how search and navigation on Helsedirektoratet's website could be improved. To provide a realistic context, the prototype loosely follows the existing visual style of *helsedirektoratet.no*. However, the project does not aim to redesign the website's layout, visual identity, or graphical elements. The primary focus is therefore on functional improvements in search and navigation rather than on visual or interface design.

The solution is intended as a conceptual prototype and is not meant to replace or be directly deployed within Helsedirektoratet's production systems. Its purpose is to illustrate how AI-assisted search and navigation could be designed and evaluated in practice. Due to limited access to internal systems and infrastructure, additional development would be required to integrate the proposed functionality into an operational environment.

While artificial intelligence is used to support improved search and navigation, the project places strong emphasis on correctness and trustworthiness of information. To reduce the risk of incorrect or misleading output, the solution avoids unconstrained generative responses and instead favors responses grounded in verified sources. The system also aims to behave consistently, meaning that identical or similar queries should result in comparable responses.

The project is further limited by computational and data-related constraints. Training large-scale AI models is outside the project scope, and the solution therefore relies on existing pre-trained models and smaller-scale experiments. Access to real user data is restricted due to privacy regulations, and opportunities for user testing are limited. As a result, user evaluations are conducted with a relatively small number of participants, which limits the breadth of empirical validation.

3.3 Product vision

The finished solution aims to demonstrate how artificial intelligence can be applied to improve search and navigation on *helsedirektoratet.no*. The primary objective is to support users in efficiently locating relevant and trustworthy information within a large and complex content structure.

By leveraging AI-based techniques such as semantic search, contextual understanding, and intelligent content categorization, the solution seeks to reduce search effort and enhance overall usability.

Rather than relying on extensive manual filtering or deep hierarchical navigation, the solution emphasizes presenting relevant information based on the user's query and contextual intent. Through improved search functionality, clearer navigation paths, and potential personalization mechanisms, the prototype illustrates how different user groups may be guided more effectively to the content most relevant to their needs.

The envisioned product is a test version of a Helsedirektoratet website, implemented as a standalone prototype. The solution consumes existing public Helsedirektoratet APIs to populate content and replicates a realistic information structure. Building on this foundation, the project introduces an AI-assisted search and navigation component designed to improve information discoverability and user experience. The prototype serves as a conceptual and experimental platform rather than a replacement for the existing production system.

3.4 Personas

To support a user-centered design process, a set of personas has been developed to represent key user groups of helsedirektoratet.no. The personas are based on anticipated user roles, information needs, and usage contexts, and are used throughout the project to guide design decisions, prioritize functionality, and evaluate proposed solutions.

The personas capture differences in professional background, goals, and challenges related to searching for and navigating health-related information. An overview of the defined personas is presented in Figure 1, Figure 2, and Figure 3.

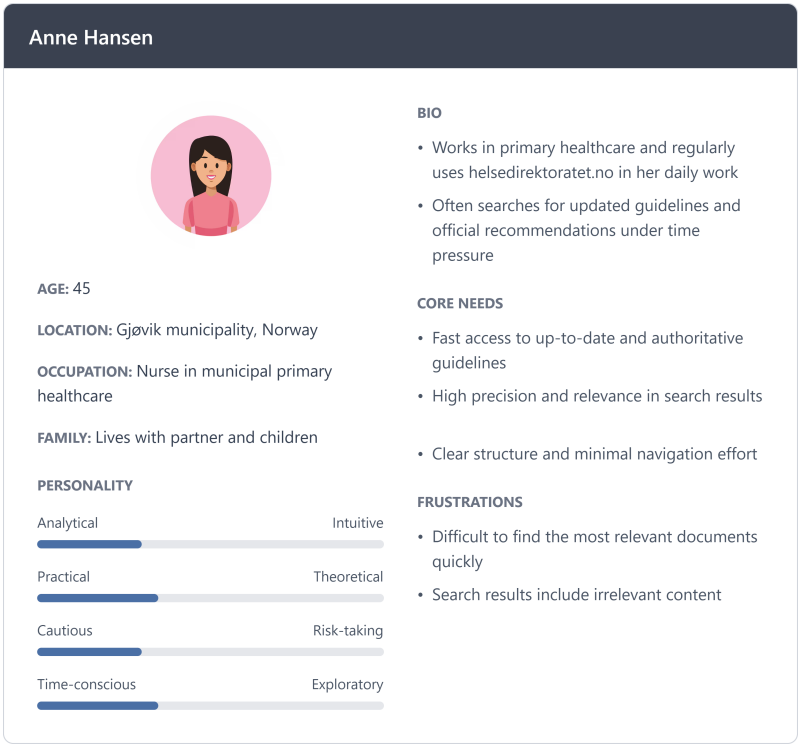


Figure 1: Persona representing a healthcare professional in municipal primary care with frequent and time-critical information needs.

Lars Nilsen

AGE: 39

LOCATION: Drammen municipality, Norway

OCCUPATION: Public health advisor in municipal administration

FAMILY: Lives with partner

PERSONALITY

Analytical	Intuitive
Practical	Theoretical
Cautious	Risk-taking
Time-conscious	Exploratory

BIO

- Works in municipal administration with public health planning and reporting
- Uses helsedirektoratet.no to find regulations, guidance documents, and national strategies relevant to municipalities

CORE NEEDS

- Overview of relevant regulations and guidance documents
- Clear structure and relationships between documents
- Easy access to updated and valid information

FRUSTRATIONS

- Difficult to understand how documents relate to each other
- Time-consuming to verify whether information is current and applicable

Figure 2: Persona representing a municipal public health advisor working with regulations, planning, and reporting.

Kari Olsen

AGE: 54

LOCATION: Oslo, Norway

OCCUPATION: HR employee (currently on medical leave)

FAMILY: Lives alone

PERSONALITY

Analytical	Intuitive
Practical	Theoretical
Cautious	Risk-taking
Time-conscious	Exploratory

BIO

- Uses helsedirektoratet.no to seek answers about her illness, treatment options, and patient rights
- Often formulates her information needs as questions and prefers step-by-step guidance
- Finds it difficult to navigate large documents and complex site structures

CORE NEEDS

- Clear and understandable answers presented in plain language
- Guidance that helps her clarify her questions and narrow down relevant information
- A supportive way of exploring information without requiring prior medical knowledge

FRUSTRATIONS

- Traditional search requires knowing specific terms
- Information is fragmented across multiple pages and documents
- Official health information is often difficult to interpret

Figure 3: Persona representing a private citizen seeking understandable and trustworthy health information.

Generic avatar icons were sourced from the Freepik icon library and used to visually support the persona descriptions. [2]

4 Project organizing

The project is organized using defined roles, responsibilities, and collaboration routines to support structured and efficient development.

4.1 Organization Map

An overview of the group structure and assigned roles within the project is presented in Figure 4.

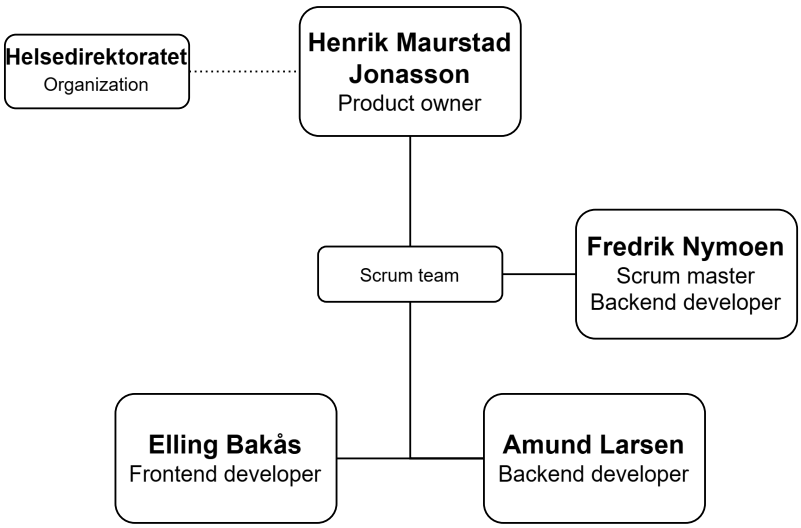


Figure 4: Organization map illustrating the project group structure and role distribution.

4.2 Responsibility and roles

The team is organized with normal scrum roles and responsiblity distrubution:

- Scrum master: Fredrik Nymoen
  - Project leader
  - Communicate with product owner
  - Schedule meetings
  - Ensure that project tasks are distributed fairly among group members
  - Set up and maintain GitHub Projects to support Scrum practices, including Kanban-style task tracking
  - Coordinate sprint planning and progress tracking
- Developer: Amund Larsen
  - Backend developer
  - Responsible for maintaining the main Overleaf report structure and ensuring clarity, consistency, and overall academic quality through review and revision.
  - Ensures that group actions and deliverables follow agreed standards and are aligned with the group contract.

- Developer: Elling Bakås
  - Frontend developer
  - Write meeting minutes from formal meetings, this includes meetings with product owner, supervisor, sprint review and sprint planning meetings.
- Product owner: Henrik Maurstad Jonasson
  - Represent Helsedirektoratet and convey their vision of the project
  - Be the contact person for the group towards Helsedirektoratet
  - Suggest ideas that could be added to the product backlog

4.3 Rules and Routines in the Group

The rules and routines governing group collaboration have been discussed and agreed upon by all group members. These are formalized in a group contract, which is included in Appendix A.

5 Team planning

Team planning is structured to support iterative development, continuous coordination, and adaptive decision-making throughout the project lifecycle.

5.1 Choice of system development model

The given task requires continuous iteration and revision throughout the project in order to adapt the resulting product to changing requirements and demands. This creates a flexible framework in which the solution can evolve over time instead of being fixed from the beginning.

The Scrum framework supports this approach by encouraging an initial idea and a high-level plan rather than a rigid, detailed plan that must be followed strictly. This flexibility allows room for feedback, learning, and improvement throughout the project lifecycle. Therefore, Scrum has been selected as the project methodology due to its agile and iterative nature. [3, pp. 37–38]

5.2 Argumentation for Scrum

Many projects involve evolving requirements, where initial assumptions about the solution may change over time. In such cases, close collaboration with the product owner and continuous feedback are essential to ensure that the solution develops in the right direction. Scrum supports this by encouraging regular communication and incremental development, allowing ideas and prototypes to be reviewed and adjusted throughout the project lifecycle.

By organizing work into short, iterative sprints, Scrum enables complex problems to be divided into manageable units. This structure simplifies planning, supports the definition of clear goals and milestones, and allows the team to continuously prioritize product backlog items based on relevance and project needs. As a result, Scrum provides an effective framework for both adaptability and structured project management. [3, pp. 39–40]

5.3 Setup of the Current Model

The project follows an adapted Agile Scrum approach with clearly defined roles mentioned earlier. The work is organized into two-week sprint cycles, providing a balance between focused development and regular opportunities for feedback, evaluation, and adjustment.

Sprint planning

Each sprint starts on a Tuesday and concludes on the following Monday. Sprint planning is conducted every other Tuesday. During this meeting, the product backlog is reviewed, a sprint backlog is established, and tasks are assigned to team members. Tasks may be refined, decomposed, or added during sprint planning to ensure that the sprint goals are realistic, well-defined, and aligned with current project priorities.

Daily Scrum

Daily scrums are held at 09:15 and are conducted physically when possible. These meetings are used to align the team on the objectives for the day, review ongoing work, and identify potential obstacles that may affect progress. The daily scrum supports continuous coordination and early detection of issues.

Supervisor meetings

Weekly supervisor meetings are held every Wednesday from 12:00 to 13:00. These meetings are used to discuss current solutions, design decisions, and ongoing challenges, and to receive guidance on the methodological aspects of the project. The supervisor meetings contribute to academic alignment and continuous reflection throughout the development process.

Sprint review

At the end of each sprint, a sprint review meeting is held every other Monday at 10:30 together with the Product Owner. The sprint review serves as the primary forum for demonstrating completed work, presenting progress relative to sprint goals, and collecting feedback. The outcome of the sprint review may lead to adjustments in priorities or scope for subsequent sprints.

Sprint retrospective

Following the sprint review, a sprint retrospective is conducted internally within the Scrum team. The retrospective focuses on reflecting on the completed sprint, evaluating collaboration and workflow, and identifying concrete improvements to be applied in the next iteration.

5.4 GitHub Projects

GitHub Projects is used as the central platform for planning, coordination, and progress tracking throughout the project. All aspects of the development process, including task management, sprint planning, meeting coordination, and milestone tracking, are integrated within a single project workspace. This provides a unified and transparent overview of both ongoing work and long-term progress.

The project is organized using multiple dedicated views tailored to different planning needs, such as a product backlog view, sprint-focused views for the current and upcoming iterations, task-specific views for individual contributors, milestone progress overviews, and roadmap views for both tasks and meetings. Together, these views function as a comprehensive Kanban-based system that supports both high-level planning and day-to-day execution.

By structuring the project in this way, GitHub Projects serves as the backbone of the team's workflow, enabling consistent tracking of tasks, responsibilities, progress, and coordination activities across the entire development lifecycle.

Workflow and Task Management

Within GitHub Projects, task flow is structured using a Kanban-based workflow that reflects the different stages of the development lifecycle. Tasks transition through clearly defined states representing planning, active development, review, completion, and potential rework.

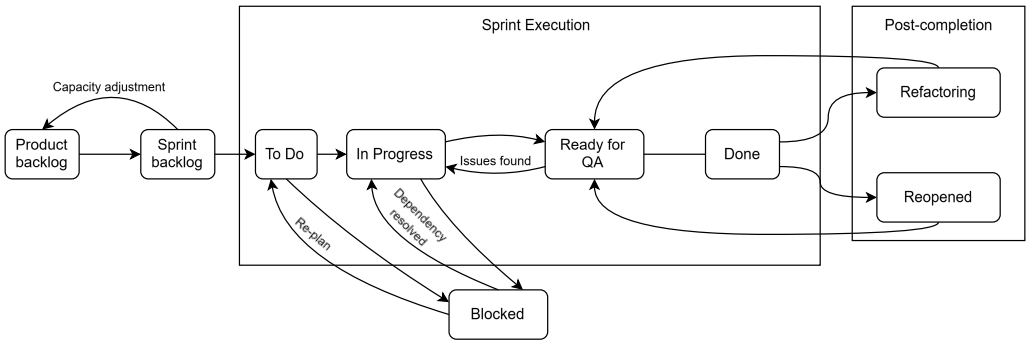


Figure 5: Development workflow and task states used in the project

As illustrated in Figure 5, this workflow provides continuous visibility into task status, workload distribution, and potential bottlenecks, enabling the team to monitor progress and react to impediments throughout each sprint.

Planning, Estimation, and Prioritization

Project work is organized around a structured product backlog, where all development tasks are defined, maintained, and tracked. Tasks can be assigned to one or more team members, ensuring clear ownership and responsibility during each sprint.

To support organization and traceability, tasks are categorized using descriptive labels that indicate their nature and purpose, such as process-related activities, implementation tasks, or milestones. Larger work items are represented as parent issues and decomposed into smaller sub-issues. Progress on parent issues is derived from the completion status of associated sub-issues, while each sub-issue maintains a reference to its parent, ensuring transparency across different levels of planning.

As illustrated in Figure 6, effort estimation is performed using a Fibonacci-based story point scale ranging from 1 to 13 instead of time-based units. Story points provide a relative measure of effort that accounts for task complexity, scope, and uncertainty, allowing estimates to remain comparable despite differences in individual working speed or implementation approach [4].

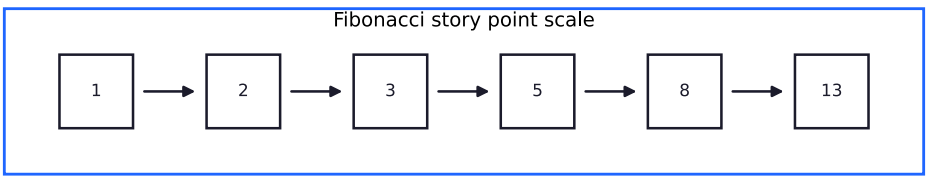


Figure 6: Fibonacci-based story point scale used for task estimation

In addition to effort estimation, each task is assigned a priority level (`P0`, `P1`, `P2`, ...), where `P0` represents the highest priority. Tasks may also be associated with a specific sprint or iteration, supporting structured planning and progress tracking (see Figure 7).

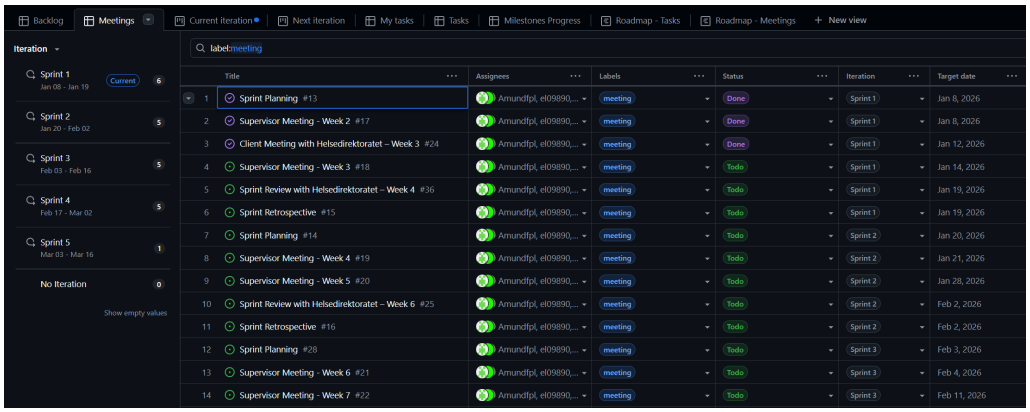
-label:meeting 31

	Title	Assignees	Labels	Sub-issues progress	Parent issue	Story Points	Status	Priority	Iteration
1	Set up github projects #10	FredrikNymoen	milestone	1 / 1 100%			Done	P0	
2	Configured GitHub Projects with boards, fields, lab... #11	FredrikNymoen	process task		Set up github proje...	3	Done	P0	Sprint 1
3	Group Contract #3	Amundfpl	milestone	0 / 1 0%			Ready for QA	P0	
4	Creating group contract #12	Amundfpl	process task		Group Contract #3	1	Ready for QA	P0	Sprint 1
5	Project Plan #2	Amundfpl, el09890,...	milestone	0 / 6 0%			In progress	P0	
6	1. Goals and Scope #4	Amundfpl	process task		Project Plan #2	1	In progress	P0	Sprint 1
7	2. Scope of the Project #5	el09890 and Fredrik...	process task		Project Plan #2	2	In progress	P0	Sprint 1
8	3. Project Organization #6	el09890	process task		Project Plan #2	2	In progress	P0	Sprint 1
9	4. Planning, Monitoring, and Reporting #7	el09890 and Fredrik...	process task		Project Plan #2	2	Ready for QA	P0	Sprint 1
10	5. Quality Assurance Organization #8	Amundfpl	process task		Project Plan #2	2	In progress	P0	Sprint 1
11	6. Implementation Plan #9		process task		Project Plan #2	2	Todo	P0	Sprint 1
12	Technical Access Clarified #37	Amundfpl, el09890,...	process			2	In progress	P0	Sprint 1
13	Current Search and Navigation Analysis #38	Amundfpl, el09890,...	process			2	In progress	P0	Sprint 1
14	User Needs and Search Challenges Identified #39	Amundfpl, el09890,...	process			8	Todo	P0	Sprint 2
15	AI Technologies and Design Alternatives Evaluated #40	Amundfpl, el09890,...	process			3	Todo	P0	Sprint 1

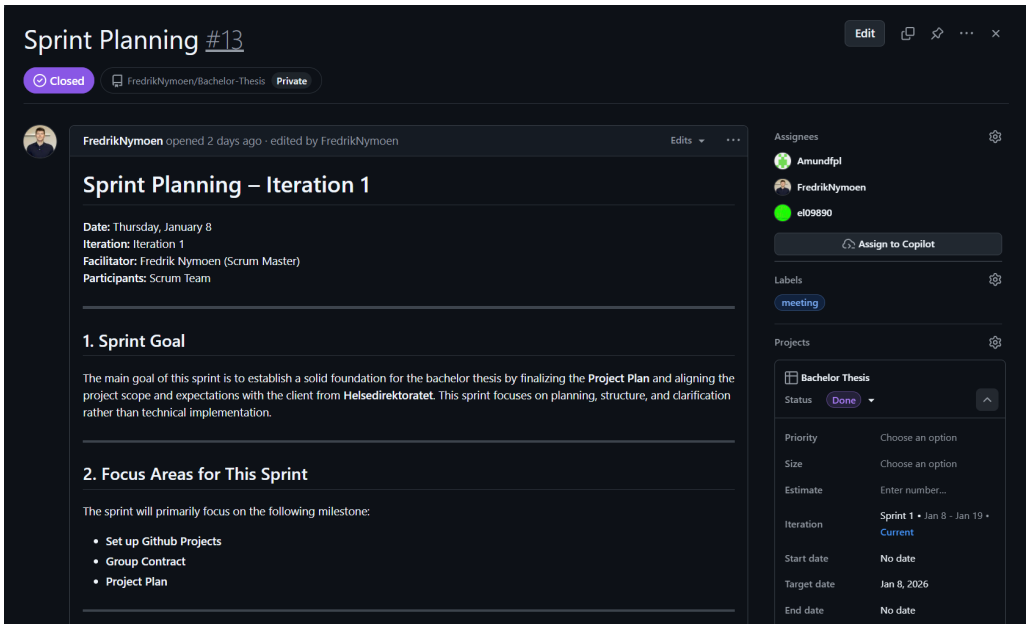
Figure 7: Overview of the GitHub Projects product backlog used for task tracking and sprint planning.

Meetings and Coordination

Meeting activities are integrated into the project workspace and managed as dedicated issues linked to the relevant sprint or iteration. This includes sprint planning sessions, sprint reviews, sprint retrospectives, as well as meetings with the product owner and academic supervisor.



(a) Meetings overview in GitHub Projects, showing scheduled and completed meetings with associated iterations and target dates.



(b) Sprint planning issue template used to document agenda items, sprint goals, and preparation notes.

Figure 8: Meeting coordination in GitHub Projects. The upper view provides an overview of meetings across iterations, while the lower view illustrates the structure of a sprint planning issue used for meeting documentation.

As shown in Figure 8, meeting issues are used to document preparation material, agendas, and outcomes such as decisions made and action items. By managing meeting planning and documentation alongside development tasks, the project maintains a centralized and traceable overview of both technical work and coordination activities, supporting continuity and transparency across sprints.

6 Quality Assurance

Quality assurance and risk management are addressed continuously throughout the project. Short sprint cycles enable early identification and mitigation of potential risks, while incremental development supports regular validation of both functionality and design decisions. Quality control is maintained throughout the development lifecycle by applying established best practices for version control, testing, documentation, and review.

6.1 Development Standards and Version Control

Version control and development standards play a central role in ensuring traceability, collaboration, and long-term maintainability. Git and GitHub are used to manage source code, documentation, and development activities across both repositories.

6.2 Architecture and Repository Structure

To support clearer separation of responsibilities and improved collaboration, the project repositories have been reorganized under a shared GitHub organization named **ai-search-navigation-bachelor**. This organization is shared among all group members and serves as the common workspace for the bachelor project.

The project is divided into multiple repositories, each with a well-defined scope. This structure enables independent development of backend and frontend components, while also providing a dedicated repository for project management and documentation. The repositories included in the organization are summarized in Table 1.

Table 1: Repository structure within the `ai-search-navigation-bachelor` organization.

Repository Name	**Description**
`helsedir-ai-backend`	Contains the backend implementation of the system, including API integrations, AI-based search logic, data processing, and supporting services.
`helsedir-ai-frontend`	Contains the frontend application responsible for user interaction, presentation of search results, and navigation features.
`Bachelor-Thesis`	Serves as the central repository for project coordination and documentation. This repository hosts the GitHub Projects (Kanban board) used for task management, as well as the LaTeX source files for the bachelor thesis, maintained under Git version control.

Branching and Code Review Practices

Development follows a structured branching and review strategy to ensure code quality, traceability, and system stability.

- Feature development is carried out on dedicated feature branches created from a shared `dev` branch.
- Feature branches are synchronized with the latest changes from the `dev` branch prior to submitting a pull request, reducing the risk of merge conflicts.
- Completed features are merged into the `dev` branch through pull requests following review.

- Stable and validated milestones are merged from the `dev` branch into the `main` branch.
- Commit messages are required to be clear, concise, and descriptive, and branch names must reflect the implemented functionality.
- Code is written according to consistent, readable, and language-appropriate coding conventions.

Issues and Traceability

GitHub Issues are used to track bugs, technical debt, and improvement proposals across both repositories. Issues are linked to relevant commits and pull requests when applicable, ensuring traceability between reported problems, implementation work, and resulting code changes. This linkage supports transparency and simplifies both review and retrospective analysis.

6.3 Testing

Testing will be an integral part of the development process and will serve as a primary mechanism for ensuring correctness, robustness, and overall solution quality. Testing activities are planned to be conducted continuously throughout development and will be adapted to the characteristics of both backend and frontend components.

Backend Testing

The backend will be implemented in Python using FastAPI. Testing will focus on verifying correct behavior of API endpoints and underlying business logic. Unit tests and API-level tests will be used to validate functionality, handle edge cases, and prevent regressions as the system evolves. Testing is planned to follow established best practices for Python-based backend development, using standard testing frameworks and tooling.

Frontend Testing

The frontend will primarily concern user interaction and visual presentation. Due to the exploratory and user-facing nature of the interface, extensive automated testing is not planned. Instead, frontend quality will be ensured through a combination of internal testing by the development team and user-oriented evaluation. Internal testing will be performed during implementation to verify expected behavior, while external testing will focus on usability, navigation, and overall user experience.

User Testing

User testing is planned to assess the usability and perceived effectiveness of both the existing helsedirektoratet.no solution and the proposed prototype. A survey-based approach will be used to collect qualitative and quantitative feedback on users' current experience, including aspects that work well and areas that present challenges.

If feasible, contact will be established with Helsedirektoratet to explore access to relevant user groups. The survey is planned to be conducted twice: once to evaluate the current solution and once after participants have interacted with the proposed prototype. The results will then be compared to assess perceived improvements in usability, navigation, and overall user experience.

6.4 Documentation Practices

Documentation is maintained alongside the source code to ensure clarity, reproducibility, and long-term maintainability. Technical documentation, architectural decisions, and usage instructions are updated continuously as the system evolves.

6.5 Tools and Technologies

The bachelor project relies on a combination of tools for analysis, prototyping, collaboration, documentation, and evaluation. These tools support the investigation of existing search and navigation solutions, the development of an AI-based prototype or concept, and the documentation of results. Table 2 summarizes the primary tools and technologies used throughout the project.

Table 2: Tools and technologies used in the bachelor project.

Tool	**Purpose**
Git & GitHub	Version control and collaborative development, including branching, change tracking, and coordination of project artifacts such as prototypes and documentation.
ChatGPT (OpenAI)	Used as a tool for debugging complex issues, refactoring code, and improving grammar and clarity in the report. The tool serves strictly as an assistant for refinement.
LaTeX (Overleaf / local)	Authoring and typesetting of the bachelor thesis, including figures, tables, diagrams, references, and structured academic formatting.
Draw.io	Creation of diagrams illustrating system architecture, search workflows, data flow, and conceptual models related to AI-based search and navigation.
Mermaid	Generation of text-based diagrams for simplified representations of system interactions, workflows, and architectural concepts.
Creately	Illustrations of personas
Microsoft Excel	Logging and tracking of individual working hours, as well as basic organization of project-related data.
Microsoft Word	Drafting meeting minutes, notes from user studies, and internal planning documentation during the project.
Discord / Microsoft Teams	Communication and coordination within the project group, including meetings, discussions, and sharing intermediate results.
Visual Studio Code	Primary development environment used for writing, editing, and maintaining the project source code.
Zotero	Reference management tool used for collecting, organizing, and citing academic and technical sources.

6.6 Risk Analysis

A risk analysis has been conducted to identify potential challenges that may affect the progress or outcome of the project. The identified risks are classified according to their *severity* and *likelihood*. Severity is categorized as *Low*, *Medium*, or *High*, reflecting the potential impact on project delivery. Likelihood is categorized as *Very unlikely*, *Unlikely*, *Possible*, or *Likely*, reflecting the probability of occurrence.

Each risk is assigned a unique identifier and is accompanied by a corresponding mitigation strategy describing how the project group plans to handle or reduce the risk. An overview of the identified project risks is presented in Table 3.

Table 3: Risk analysis with severity, likelihood, and handling status.

ID	Risk Description	Severity	Likelihood	Status
R1	Project falls behind schedule, leading to delayed or incomplete delivery of the final report or prototype.	High	Possible	Handled
R2	Long-term illness or extended absence of one or more group members.	High	Unlikely	Handled
R3	Prolonged inactivity or lack of communication from a group member critical to project progress.	High	Unlikely	Handled
R4	Changes in project scope or expectations from the product owner or supervisor.	Low	Likely	Handled
R5	Technical challenges related to AI tools or data availability limit the feasibility of the proposed solution.	Medium	Possible	Accepted
R6	Helsedirektoratets API becomes unavailable.	Medium	Unlikely	Handled
R7	Overleaf servers become unavailable, resulting in temporary or permanent loss of access to the project report.	High	Possible	Handled
R8	The group loses access to Microsoft Azure, resulting in the service becoming unavailable.	Medium	Possible	Handled
R9	Limited ability to collect sufficient user feedback to accurately assess the current user experience of *Helsedirektoratet.no* and to validate the proposed solution.	Medium	Possible	Handled
R10	The prototype may not accurately reflect how real users expect search and navigation on helsedirektoratet.no to behave, limiting the validity of evaluation results.	Medium	Possible	Handled
R11	AI-based techniques may unintentionally favor certain document types, topics, or terminology, leading to biased or incomplete search results.	Medium	Possible	Handled
R12	Users may not understand why certain results are ranked or suggested, potentially reducing trust in the system.	Medium	Possible	Handled
R13	Findings from limited user testing may not be generalizable to all target groups of helsedirektoratet.no.	Low	Likely	Accepted
R14	Changes in availability, pricing, or behavior of third-party AI models or libraries could affect prototype functionality.	Medium	Unlikely	Handled

Risks marked as *Handled* in Table 3 have corresponding mitigation strategies described in Section 6.7 and summarized in Table 4, while risks marked as *Accepted* are monitored but not actively mitigated.

6.7 Plan for Handling Identified Risks

Table 4: Countermeasures for identified project risks.

Risk ID	**Planned Countermeasures**
R1	Define clear milestones and internal deadlines, conduct regular progress reviews, and reallocate tasks if delays occur. Progress is monitored continuously to ensure timely delivery of both the prototype and final report.
R2	Ensure thorough knowledge sharing across group members through shared documentation and regular status updates. In the event of prolonged absence, responsibilities will be redistributed to maintain project continuity.
R3	Follow the escalation procedure defined in the collaboration agreement, including internal discussions and, if necessary, involvement of the project supervisor to resolve communication or participation issues.
R4	Maintain regular communication with the product owner or supervisor, document key decisions, and manage scope changes by prioritizing core project objectives and adjusting secondary features if required.
R6	Contact Helsedirektoratet and find out the estimated downtime length. If downtime is estimated to be short, then work can continue working with other things. If downtime is estimated to be long then a mock API could be created
R7	The project report is maintained under Git version control with a mirrored local and remote repository. This ensures continuous access to all report versions and prevents data loss in the event of Overleaf service outages or platform unavailability.
R8	The source code is managed using Git version control, and the project is containerized with Docker, enabling a smooth and rapid transition to an alternative hosting provider if required.
R9	Mitigated by clearly defining the scope and limitations of the user studies and by using the feedback that is available as qualitative guidance when refining the solution. Findings will be interpreted in light of the limited sample size and treated as indicative rather than representative.
R10	Use personas, realistic test tasks, and scenario-based evaluation to align prototype behavior with real-world usage patterns. Clearly document assumptions and limitations.
R11	Limit AI output to verified content sources, evaluate result diversity during testing, and document observed biases as part of the evaluation.
R12	Include explainability cues (e.g., source references, ranking rationale where feasible) and explicitly discuss ethical considerations and transparency limitations in the thesis.
R14	Rely on stable, well-documented libraries and design the prototype so AI components can be replaced or mocked if necessary.

7 Implementation

This section outlines the planned implementation approach for the project, including key activities, technical focus areas, and intended development steps. The purpose is to provide an overview of how the proposed solution will be realized over time and to establish a reference for prioritization and progress tracking during the project.

7.1 Planned System Implementation

The implementation of the proposed solution is planned to proceed through a series of structured steps, reflecting both technical development and user-centered considerations. The following activities represent the primary implementation goals for the project:

- Collect and analyze user feedback through surveys to understand current search behavior, navigation patterns, and user challenges.
- Identify and make use of publicly available data and APIs provided by helsedirektoratet.no to populate the prototype with realistic and relevant content.
- Develop a frontend prototype that resembles the structure and visual style of helsedirektoratet.no, while incorporating proposed improvements to search and navigation.
- If necessary, store selected data in a dedicated database to support preprocessing, experimentation, and efficient data management for AI-based components.
- Implement a backend service that integrates improved search and navigation functionality, serving as the core logic of the proposed solution.
- Investigate automated labeling and tagging of content using natural language processing techniques to enhance search relevance and content organization.
- Assess the feasibility of enhancing the search interface with interactive elements, such as query refinement or guided search, provided that time and technical constraints allow.

The listed implementation steps are intended to guide development and prioritization throughout the project. Depending on time constraints, technical feasibility, and evaluation needs, the scope and order of these activities may be adjusted to ensure that the core project objectives are achieved.

7.2 Gantt diagram

The project timeline and planned distribution of activities are illustrated in Figure 9. The Gantt diagram provides an overview of the main project phases, key milestones, and their expected duration over the project period. It serves as a planning and coordination tool to ensure that dependencies between tasks are identified and that progress can be monitored against the planned schedule.

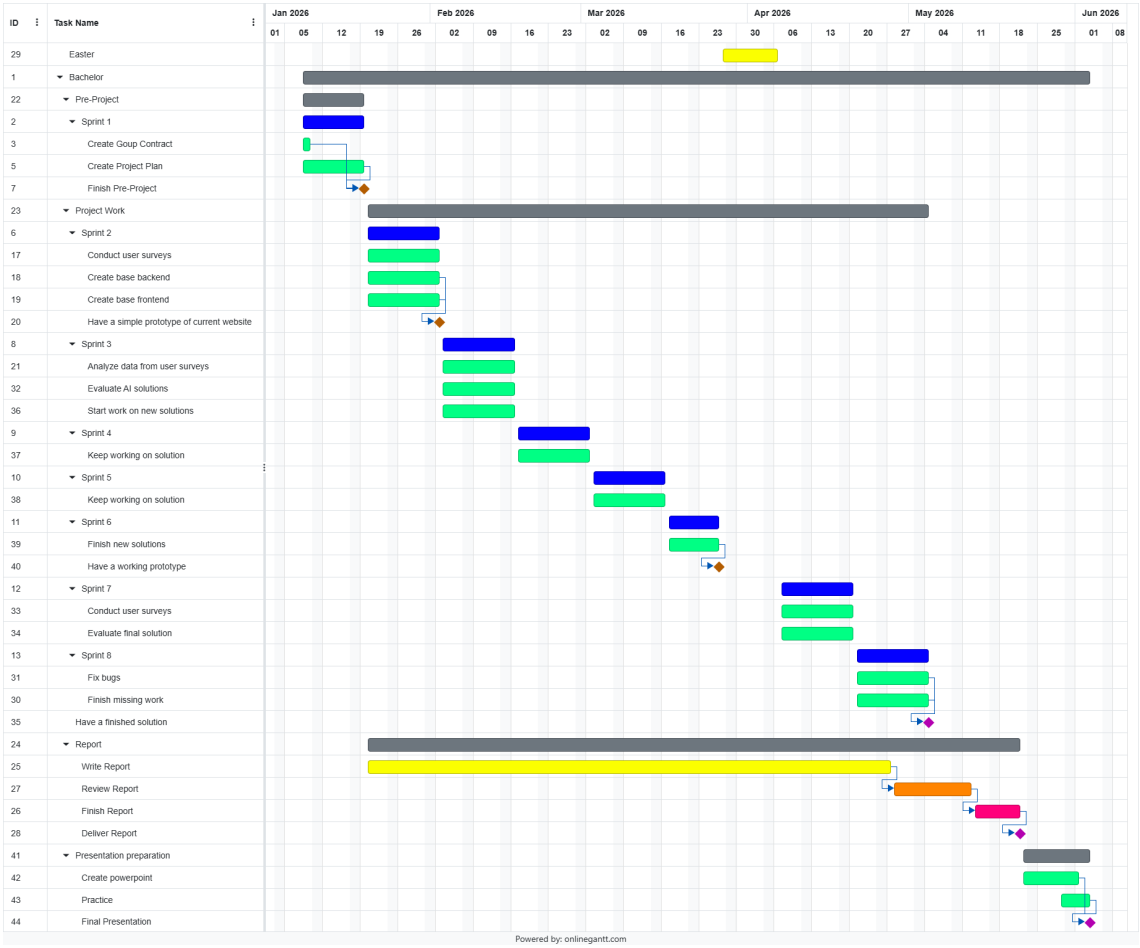


Figure 9: Gantt diagram

7.3 Progress Plan

Pre-Project Phase: Weeks 2 - 3

- Establish contact with supervisor and product owner
- Create group contract
- Create project plan

Main project work: Weeks 4 - 17

- Conduct user studies and analyze current search methodology
- Create a website that imitates the Helsedirektoratets website
- Create a backend that communicates with our webpage and Helsedirektoratets APIs
- Experiment with new solutions for navigation on the webpage
- Conduct user studies to evaluate our solution
- Write report
- Ducument the process

Finishing work: Weeks 18 - 21

- Focus on finishing the report
- Test and bugfix the application

Presentation preparation: Weeks 21 - 23

- Create a Powerpoint presentation
- Practice for the presentation

References

[1] Helsedirektoratet. *Om Helsedirektoratet*. URL: https://www.helsedirektoratet.no/om-oss (visited on Jan. 15, 2026).

[2] Freepik Company. *Freepik Icon Library*. URL: https://www.freepik.com/icons (visited on Jan. 16, 2026).

[3] Ian Sommerville. *Engineering Software Products: An Introduction to Modern Software Engineering*. Global Edition. Harlow, England: Pearson Education, 2021. ISBN: 978-1-292-37634-9.

[4] Wrike. *The Complete Guide to Scrum Story Points*. 2023. URL: https://www.wrike.com/scrum-guide/faq/what-are-scrum-story-points/ (visited on Jan. 13, 2026).

Appendix

A Group Contract

The group contract is attached on the following page(s).

Collaboration Agreement

The following collaboration agreement defines the expectations, responsibilities, and procedures agreed upon by all group members for the duration of the project.

1. Each group member is expected to contribute approximately 30 hours of project work per week on average.
2. The group will hold a minimum of two physical meetings per week, with a goal of meeting up to four times per week when feasible.
3. Each group member is responsible for registering their working hours accurately in the shared timesheet.
4. In the event of absence, a group member must notify the rest of the group as early as possible, and preferably at least 24 hours in advance.
5. If a group member anticipates being late to a scheduled meeting, prior notice must be given to the rest of the group.
6. *Fredrik Emil Nymoen* serves as a working project leader. This role includes responsibility for scheduling meetings, maintaining the collaboration agreement, ensuring that project tasks are distributed fairly among group members, and overseeing communication with the client.
7. All group members share responsibility for reviewing project deliverables to ensure that the work meets the required quality standards.
8. Failure to comply with the obligations outlined in this agreement may result in the initiation of the procedure described in Section "No Show and Complications"

No Show and Complications

In cases where a group member repeatedly fails to meet agreed-upon responsibilities, the following escalation procedure shall be followed:

1. An internal discussion involving all group members to address the issue and seek a resolution.
2. A written warning issued by the remaining group members, which shall include:
   - A clear description of the violated agreement or obligation.
   - Specification of any required compensation for lost or redistributed workload.
   - Clarification of potential consequences in the event of further violations.
3. A formal meeting involving the entire group and the project supervisor.
4. If the matter remains unresolved, the subject coordinator shall be involved to assist in finding an appropriate solution.
5. As a final measure, a decision on written exclusion from the group may be made, subject to approval by the remaining group members, the project supervisor, and the subject coordinator.

Group Members This collaboration agreement applies to the following group members:

- Fredrik Nymoen
- Elling Bakås
- Amund Finne Præsteng Larsen